

WELCOME TO JOSH JAMES DJ TRAINING!

This course is designed to give you all the tools you need to become a competent, confident professional DJ.

Understanding the equipment and how it works together is a key skill set in becoming a great DJ, although the equipment we will be using in the lessons is Pioneer, all of the following information can be used across any make and model of mixer, player or controller.

You will learn the fundamental musical elements behind every great mix and how to adapt them to your own personal style. There will also be heaps of handy bonus tips along the way!

It is important to continue to seek further education! The skills and techniques in this course are core foundational tools and everything you need to become a pro. Once you master them, there are infinite ways to create amazing mixes. Take inspiration from everywhere you can and explore your creativity to discover exciting new ways to mix.

This course will be as broad as possible when it comes to equipment specifics. I encourage all students to be resourceful when having any technical issues, one of the best skills I ever developed was the ability to figure things out on my own. Go online, youtube, Reddit.. anywhere!! There has definitely been someone out there with the same issue you are having now.

Join our Facebook group to connect with other students, share mixes and gigs, and give each other help and support.

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- **IEC** and **USB** are used to supply power.
- **TRS, RCA, XLR** all carry **audio signals**, they do the same thing, they just have different connections.
- At least two of these leads/connectors will be used in a set-up, getting to know what each one does will ensure you are prepared to fix any issues that may arise before or during a gig.
- Pride yourself on knowledge, the more you know about your craft inside and out makes you not just good, it makes you a master.

Signal Flow

- **IEC** connections run to a wall socket, these connections are sturdy and reliable. I recommend using one when given the option.
- **USB** supplies data and in the case of using software with a controller it can also be a power supply.
- **XLR** and **RCA** are most commonly used for the **output** signal going to the speakers. **TRS** connections are also used for output signals however this often goes to foldback or reference speakers.
- Now we are all set up, take this time to make sure your equipment is running properly. If you're having any issues with connecting your equipment now is the perfect time to do your first bit of research... Head to youtube or the facebook group and figure it out!

Gain

- **Gain** is another expression for volume.
- Setting the equipment **output gain** and performing a **soundcheck** before starting the mix is a must! We want to make sure all the amazing transitions we are pulling off can be heard. Make sure the **output gain** of the equipment has been set and tested.



- On the speaker it is good to have the **volume pot** set to half or greater this will achieve great sound production from the speaker.
- Adjust the **mixer** and **channel gain pots** to set the **output gain** heading to the speakers, avoiding pushing the **output gain** too much. Use the **gain indication lights** on the mixer to assist in achieving balance and avoid making one or the other work too hard.

MODULE 2: PLAYER BASIC FUNCTIONS

Tempo Functions

- Use the **Tempo Slider** to set the **BPM** or “*speed*” of the track, sliding it up will decrease and sliding it down will increase. Return it to the middle to reset the track to its original **BPM**.

#TIP: ALWAYS CHECK BOTH YOUR TRACKS BPM MATCH BEFORE STARTING TO MIX!

- The **Tempo Range Selector** sets the amount the **Tempo Slider** can adjust, the amounts are
 - +6 (smallest range)
 - +10 (most common)
 - +16 (common in hip hop & RnB)
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- **Master Tempo** This button prevents the pitch of the track from changing when adjusting the tempo. I recommend keeping it on at all times by ensuring the **red light** is on. If you don't have a physical button on your equipment, look for the **MT** symbol on the display.
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Jog Wheel

- The **Jog Wheel** is used to move the track forwards and backwards while it is playing; this is how you align the two tracks when **beat matching**.
- This wheel is touch-sensitive and when set to **Vinyl Mode** will act in the same way a record does allow you to scratch and stop the track by pushing down on the wheel.

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CUE/Play

- It is important to fully understand these buttons and their purpose. It may seem obvious but these buttons really are the most important to master.
- **PLAY/PAUSE** the **green** button (in most cases) does exactly that, it Plays the track press it once and off you go. Press it again and it will pause the track, use this in conjunction with the cue button to set new cue points.

****note; A Second push will pause the track where it is pressed, if you wish to restart the song from the original cue point press the CUE button.***

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Cue audition/Headphone controls

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- Faders control the volume of each track, the two vertical **Faders** are assigned to the channels labelled above the faders.
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****note; It's completely up to you, whatever fader fits your style of mixing best!***

- Channel faders do not always need to be pushed all the way to 10! Spend some time learning the relationship between fade level and volume, some things sound better when they are quiet.

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- The EQ pots control the frequency output on a channel, they are split into three “bands” **Hi/Mid/Low** and can usually increase or decrease a frequency band in volume by about half.
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- **Sound Colour Fx** are channel affecting only and can not be used on the master channel. There are usually at least two effect options, highlighted by white light in this case and will flash when activated. The **Parameter** pot controls the overall intensity of the effect.
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- In the Beat FX section, there are a few ways to set the cycle duration on most equipment, in this example, the two arrows labelled **BEAT** will change the duration of the cycle up or down, for example, 1 beat down to $\frac{1}{2}$ beat.
- Also commonly found in the **Tap Tempo** button which allows you to set the tempo of the **Beat Fx** with your current track selection by tapping it in time this will keep your effect in time with your track.
- You can also set the duration in smaller increments with the **Time** pot found on some mixers and controllers.
- Use the **Fx Select** pot to choose your desired effect; most mixers and controllers have a bank of at least 4 effects; these often include **reverb**, **phaser**, **echo** and **pitch**.

****Note; Software and controller setups may have Beat FX controlled in the software if this is the case do a bit of research and figure out how they are used on your specific setup.***

- The **Fx Designator** pot that sets what the source the FX will be applied to eg. channel 1 or master. The **Level/Depth** pot controls the intensity of the FX and the blue button underneath turns the FX on and off.

MODULE 4 UNDERSTANDING MUSIC

BPM

- **BPM** stands for **beats per minute**; it is the **tempo** or “*speed*” of the track.
- While you are learning, stick to mixing music that is similar in tempo. For example; house and electronic tracks in the range of 120bpm - 128bpm. On hip hop tracks in some cases, adjustments can be made over a greater range but I still recommend sticking to tracks with the same or similar **BPM** while you learn.

Key

- The **Key** is the musical note or chord the track is centred around, for example, tracks in the **Key** of Cmaj would be centred around a **C note**. There are two different ways Keys will be displayed, either the traditional **C Bm** or alphanumeric combos **4A 8B**.
note for a general rule of thumb **Major Keys go with **Major Keys** and **Minor** with **Minor**. With numbered letters, it's much easier! Just one part has to match ei. **4B** to **4A** to **7A** to **7B** to **1B**. When the numbers match up, that's the **relative key**, a **key change** from **minor** to **major** or vice versa.*
- This type of key coding is commonly used for its “easy to use” approach to mixing in keys. Why is this important? Just like the notes used in the song, if the key of the selected tracks aren't in tune they can sound unpleasant or off when mixed together.
****note; a good mix is not only in time and beat-matched it is harmonically please also.***

Structure

- To become a good DJ you must be able to count to 32... In most cases electronic music genres such as house, trance, drum n bass, hip hop and r&b. will all follow the same constructional pattern, beats and bars and phrases. 4 beats to a bar, 8 bars to a phrase.
- In the production of electronic music, the general practice is to add something in or take something away every 8 bars. You can use this same pattern to make your mixes sound natural.
- The structure of these genres will usually be composed of these sections;

INTO/VERSE → DROP/CHORUS → BREAKDOWN/BRIDGE → OUTRO

TRY THIS:

Put a track on and count along with the kick drum, and listen to the new elements introduced. Notice that on or around every 32 beats something new happens in the track.

#PRO TIP:

This is the golden ticket, the sauce, the one skill that allows you to dominate every single mix... Get your head around the structure of music! If you master this you can mix absolutely anything!

MODULE 5 MIXING TECHNIQUES

What is mixing?

Mixing is the practice of transitioning seamlessly between multiple tracks creating a listening experience that is dynamic and exciting to the listener. Well, it should be anyway... Think of your goal as trying to make two songs sound like one through good timing and energy transfer.

Counting beats and bars

- Get used to counting the beats while you are playing.. Actually, Anytime you hear music. Make it a habit! The more often you do this the sooner you will start to do it subconsciously, leaving you more brain juice to use on crazy effects and wicked good transitions.
- When counting in, remember we are looking for the start of the **phrase**. To recap on the structure lesson in the previous module; 4 beats to a bar, 8 bars to a phrase.
- The new phrase is always marked by something changing in the song whether it's a new hi-hat line in the drums or a full track breakdown; it happens at the start of a phrase... every 32 beats.

Beat matching

- Beat matching is exactly that, matching the beats. Do this by lining up the kick drums or beats directly over one another.
- At some point, you will come across a situation when you don't have a kick to line up, in this case, listen for any drums you can keep time with. For now, I recommend sticking to tracks that start with a kick drum while learning.
- Starting a track early or late by a fraction will be noticeable and unpleasant and must be aligned before being introduced to the mix.
- To do this move the jog wheel counterclockwise to slow the track or clockwise to speed it up.

**** note; practice makes perfect on this one, you will need to do this many times to master it!***

Timing

- Everything you do in a mix must be in time! The new track must be played in time, the volume of that track must be adjusted in time, the frequencies must be transferred in time. If you are doing anything in a mix, do it in time. Every 8 beats you can do a part of the process.
- A good place to start your counting is the second drop, this is the start of the phrase! From the drop count to 32 and hit play on 1, both tracks are now playing in time with each other. Being in time means that the first beat of the phrase on both tracks are in line, this is because the play was pressed at the start of the phrase.
- When starting a new mix play is always pressed on the first beat of a new phrase

Transitions

- **Drop mixing** is a transition method where the next track is “dropped” in or swapped immediately with the current track, this is usually done in the drop or somewhere the current track loses energy. For example in the **breakdown** when the bass and drums stop.

#PRO TIP:

When performing a drop mix choose a section of the song that you want to drop in, if the song doesn't have an intro then you can start at that point otherwise make a cue point in the track at the point you would like to drop it in.

- Alternatively, you can **beat match** the song with the intro as usual without bringing the volume in, then wait for the point where you want to swap the tracks and just do a simple volume swap.

****note review the lesson on structure for a recap on breakdowns and drops***

- **Frequency transfer** is a transition method involving the swapping of frequency bands in time to blend the two tracks together over time. An example of this is;
 - 1. **1st beat** - hit play
 - 8 beats** - beat match
 - 16 beats** - track volume up with no low band.
 - 24 beats** - swap low bands.
 - 32 beats** - pull the previous track out.

****note: It is important that every element of your mix is done in time to make the transition feel as natural as possible.***

Energy

- Energy is one of the most important elements in any DJ mix, all energy transfers should be done in time. If too much energy is added at the wrong point in a mix it will sound unnatural and noticeable, the same goes for energy that is taken out at the wrong point. Remember the point of a good mix is to be seamless, smooth, almost inaudible to the listener.

#PRO TIP:

Ask yourself, am I moving the energy in the right direction, Visualise your set as graph over time. See the energy moving over time, try to relate with your listeners and how it will move them. Not all sets will necessarily have the same energy flow, some will need a constant level of energy others will need to rise and fall.

Looping

- Sometimes you might find that one of your tracks only has an 8 beat intro or that you only have 15 seconds left on the currently playing track, setting a beat loop will give the extra time needed to pull off a good clean mix.
- Looping can also be used to sample certain parts of a track and mix them in with the current track. For example, setting a 16 bar loop in the vocal section of your second track and mixing that into the currently playing track to create your own remix live!

Effects

- Using the built-in effects on the mixer is another way to make your mix exciting and dynamic. Adding things such as Delay or Echo in combination with a high pass filter is one way to spice up an otherwise basic frequency transition or drop mix.

GLOSSARY

Audio Signals - Audio signals are the representation of sound, which is in the form of digital and analog signals. Their frequencies range between 20 to 20,000 Hz, which is our ears' lower and upper limit. Analog signals occur in electrical signals, while digital signals occur in binary representations

Gain - the amount of input you give a system

IEC - used as power connectors for electrical appliances. IEC refers to the set of standards from the International Electrotechnical Commission.

Output Gain - to increase the power or amplitude of a signal from the input to the output

Soundcheck - the preparation that takes place before a concert, speech, or similar performance to adjust the sound on the venue's sound reinforcement or public address system.

RCA - a type of electrical connector commonly used to carry audio and video signals. The name *RCA* derives from the company Radio Corporation of America.

TRS - a common audio connector and stands for Tip, Ring, Sleeve. It is cylindrical in shape, typically with three contacts, although sometimes with two (a TS connector) or four (a TRRS connector).

USB - is an industry-standard cable and connector for communication and power supply between computers and other electronic equipment. A broad variety of USB hardware exists, including 14 different connector types, of which USB-C is the most recent.

XLR - a type of electrical connector primarily found in professional audio, video, and stage lighting equipment. The connectors are circular in design and have between three and seven pins.

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MODULE 5 MIXING TECHNIQUES

What is mixing?

Mixing is the practice of transitioning seamlessly between multiple tracks creating a listening experience that is dynamic and exciting to the listener. Well, it should be anyway... Think of your goal as trying to make two songs sound like one through good timing and energy transfer.

Counting beats and bars

- Get used to counting the beats while you are playing.. Actually, Anytime you hear music. Make it a habit! The more often you do this the sooner you will start to do it subconsciously, leaving you more brain juice to use on crazy effects and wicked good transitions.
- When counting in, remember we are looking for the start of the **phrase**. To recap on the structure lesson in the previous module; 4 beats to a bar, 8 bars to a phrase.
- The new phrase is always marked by something changing in the song whether it's a new hi-hat line in the drums or a full track breakdown; it happens at the start of a phrase... every 32 beats.

Beat matching

- Beat matching is exactly that, matching the beats. Do this by lining up the kick drums or beats directly over one another.
- At some point, you will come across a situation when you don't have a kick to line up, in this case, listen for any drums you can keep time with. For now, I recommend sticking to tracks that start with a kick drum while learning.
- Starting a track early or late by a fraction will be noticeable and unpleasant and must be aligned before being introduced to the mix.
- To do this move the jog wheel counterclockwise to slow the track or clockwise to speed it up.

**** note; practice makes perfect on this one, you will need to do this many times to master it!***

Timing

- Everything you do in a mix must be in time! The new track must be played in time, the volume of that track must be adjusted in time, the frequencies must be transferred in time. If you are doing anything in a mix, do it in time. Every 8 beats you can do a part of the process.
- A good place to start your counting is the second drop, this is the start of the phrase! From the drop count to 32 and hit play on 1, both tracks are now playing in time with each other. Being in time means that the first beat of the phrase on both tracks are in line, this is because the play was pressed at the start of the phrase.
- When starting a new mix play is always pressed on the first beat of a new phrase

Transitions

- **Drop mixing** is a transition method where the next track is “dropped” in or swapped immediately with the current track, this is usually done in the drop or somewhere the current track loses energy. For example in the **breakdown** when the bass and drums stop.

#PRO TIP:

When performing a drop mix choose a section of the song that you want to drop in, if the song doesn't have an intro then you can start at that point otherwise make a cue point in the track at the point you would like to drop it in.

- Alternatively, you can **beat match** the song with the intro as usual without bringing the volume in then wait for the point where you want to swap the tracks and just do a simple volume swap.

****note review the lesson on structure for a recap on breakdowns and drops***

- **Frequency transfer** is a transition method involving the swapping of frequency bands in time to blend the two tracks together over time. An example of this is;
 1. **1st beat** - hit play
 - 8 beats** - beat match
 - 16 beats** - track volume up with no low band.
 - 24 beats** - swap low bands.
 - 32 beats** - pull the previous track out.

****note: It is important that every element of your mix is done in time to make the transition feel as natural as possible.***

Energy

- Energy is one of the most important elements in any DJ mix, all energy transfers should be done in time. If too much energy is added at the wrong point in a mix it will sound unnatural and noticeable, the same goes for energy that is taken out at the wrong point. Remember the point of a good mix is to be seamless, smooth, almost inaudible to the listener.

#PRO TIP:

Ask yourself, am I moving the energy in the right direction, Visualise your set as graph over time. See the energy moving over time, try to relate with your listeners and how it will move them. Not all sets will necessarily have the same energy flow, some will need a constant level of energy others will need to rise and fall.

Looping

- Sometimes you might find that one of your tracks only has an 8 beat intro or that you only have 15 seconds left on the currently playing track, setting a beat loop will give the extra time needed to pull off a good clean mix.
- Looping can also be used to sample certain parts of a track and mix them in with the current track. For example, setting a 16 bar loop in the vocal section of your second track and mixing that into the currently playing track to create your own remix live!

Effects

- Using the built-in effects on the mixer is another way to make your mix exciting and dynamic. Adding things such as Delay or Echo in combination with a high pass filter is one way to spice up an otherwise basic frequency transition or drop mix.

GLOSSARY

Audio Signals - Audio signals are the representation of sound, which is in the form of digital and analog signals. Their frequencies range between 20 to 20,000 Hz, which is our ears' lower and upper limit. Analog signals occur in electrical signals, while digital signals occur in binary representations

Gain - the amount of input you give a system

IEC - used as power connectors for electrical appliances. IEC refers to the set of standards from the International Electrotechnical Commission.

Output Gain - to increase the power or amplitude of a signal from the input to the output

Soundcheck - the preparation that takes place before a concert, speech, or similar performance to adjust the sound on the venue's sound reinforcement or public address system.

RCA - a type of electrical connector commonly used to carry audio and video signals. The name *RCA* derives from the company Radio Corporation of America.

TRS - a common audio connector and stands for Tip, Ring, Sleeve. It is cylindrical in shape, typically with three contacts, although sometimes with two (a TS connector) or four (a TRRS connector).

USB - is an industry-standard cable and connector for communication and power supply between computers and other electronic equipment. A broad variety of USB hardware exists, including 14 different connector types, of which USB-C is the most recent.

XLR - a type of electrical connector primarily found in professional audio, video, and stage lighting equipment. The connectors are circular in design and have between three and seven pins.